Sahaj Bhadja

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Summary

Agentic AI Engineer with 2+ years building production Python and LLM systems, including **AI agents**, **RAG**, **workflow orchestration**, evaluation, governance, and stakeholder-led cloud delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Advanced NLP, Database Systems, Analysis of Algorithms, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Data Retrieval, Operating Systems, Object-Oriented Programming

Experience

USC Dornsife **September 2025 – Present**
Forward Deployed Software Engineer *Los Angeles, CA*

- Translated stakeholder workflows into **production LLM orchestration** with FastAPI workers, explicit job states, failure isolation, and monitoring, processing 1K+ AI jobs at 3x throughput without cascading failures under concurrent load.
- Built multi-tenant data and integration pipelines across Azure SQL, PostgreSQL, SQL Server, and cloud storage, processing 5K+ monthly jobs with **validation, reconciliation, CI/CD, and 99%+ reliability** across production environments.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built a **unified AI model layer** across OpenAI, Anthropic, and Qwen with multimodal routing, fallbacks, timeout handling, and usage tracking, serving 10K+ daily requests for 150+ instructors.
- Converted instructor requirements into React and Flask workflows for uploads, rubrics, grading, and feedback, reducing turnaround 60% while validating releases through **automated tests, reviews, and production monitoring**.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built a GPT-based Decision Transformer in **PyTorch** with evaluation pipelines over 100K+ trajectory logs, improving long-horizon planning 18% while documenting model behavior, experiments, and measurable results for autonomous research workflows.
- Defined repeatable **evaluation frameworks** across three planning baselines, using controlled simulations and experiment tracking to assess accuracy, failure modes, and improvements that increased closed-loop stability 15% for reliable research decisions.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Translated requirements into object-oriented microservices and large-scale API integrations, improving response times 30% and maintainability while collaborating with Microsoft and GitHub stakeholders through **Agile delivery and technical reviews**.
- Automated **quality gates and release evidence** with 20+ PyTest and Mockito suites, Jenkins pipelines, VM builds, and cross-platform validation, increasing pipeline efficiency 68% and reducing technical debt 20%.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Built eight modular microservices for authentication, permissions, resources, billing, and notifications, using **REST APIs and event-driven WebSockets** to synchronize 2M+ records across distributed locations through secure backend services.
- Developed React and TypeScript interfaces with real-time dashboards, then supported production releases through Docker staging, Jenkins CI/CD, Sentry monitoring, and **documented operational troubleshooting** across eight services for reliable operations.

Skills

Languages: Python, TypeScript, JavaScript, Java, SQL
Agentic AI: AI Agents, Large Language Models, Prompt Engineering, RAG, Tool Integration, Workflow Orchestration, Embeddings, FAISS, Pinecone, PyTorch, Transformers, Multimodal AI
Backend & Data: FastAPI, Flask, REST APIs, Microservices, PostgreSQL, SQL Server, MySQL, Data Modeling, ETL
Cloud & Operations: Azure, AWS, Docker, Kubernetes, CI/CD, Jenkins, GitHub Actions, Monitoring, Logging, Sentry
Delivery & Quality: Git, Jira, Agile, Requirements Analysis, Technical Specifications, Evaluation Frameworks, Model Validation, Unit Testing, Integration Testing, Documentation, Stakeholder Management

Projects

AgentOps Studio | *Python, FastAPI, PostgreSQL, Redis, RAG, Tool Integration, Docker*

- Designed an **agentic workflow platform** that converts operational requests into scoped plans, retrieves governed enterprise context, invokes approved tools, records human approvals, and exposes auditable execution states through FastAPI services.

EvalGuard | *Python, PyTest, Embeddings, PostgreSQL, GitHub Actions, Monitoring*

- Built an **LLM evaluation and monitoring framework** with curated test cases, retrieval-grounding checks, tool-call validation, regression thresholds, and review dashboards supporting repeatable release decisions and continuous agent improvement.

Big-Brain | *LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP, REST APIs* | Live

- Built an **AI application** with LLaMA-2, transformer embeddings, and FAISS semantic search over 50K+ documents with Clerk authentication, cutting retrieval latency 40% at 95% query accuracy in production.